

RENANCE

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

CHM 223 — Past Questions & Materials
by Renance

Course: CHM 223
Institution: Federal University of Petroleum Resources, Effurun
Papers in this pack: 3
Levels: 200

This study pack was compiled and restructured by Renance for CHM 223 students at Federal University of Petroleum Resources, Effurun. It brings together 3 past paper(s)/material(s) into one clean, readable document so you can revise without chasing scattered PDFs. Practise these same questions interactively on Renance with instant answers, AI explanations, and full CBT mock exams — over 235,000+ questions across JAMB, Post-UTME, WAEC, NECO and university courses.

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Paper 1 of 3 — CHM 223, 2024 past question

Level 200 • Second Semester • PastQuestion

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DEPARTMENT OF CHEMISTRY ,

SECOND apace en PUYSICAL CHEMISTRY HH (2 UNITS)

INSTRUCTIONS: aienn as Lge EXAMINA) 'GN; 2023/2024 ACADEMIC SESSION

PANO WER QUESTION 1 AND ANY OTHER FOUR QUESTIONS TIME: 2 HOURS

DELTA STATE

I. Fee examples cach, explain inte

(splain the fe 'ing te ss ot . : a :

'sob "[: ne following terms: adiabatic process, isochoric process, isothermal process, tsobaric process and independent state variables

()- . ae te the value of AE and ALL on heatins 64 g of Oz from 0 °C to 100 °C. Cv and Cp on an average are 5.0 and 7.0 cal. mol⁻¹°C⁻¹ :

2. (a) Consider a gas expanding isothermally from 1 dm³ to 2 dm³ at 300 K. calculate the change in entropy

if the pressure changes accordingly. :

(b) Fora substance with a latent heat Of vaporization of 2250 J/g, find the change in entropy when 100g

of the Substance is vaporized at its boiling point of 373 K. (c) A gas with a constant heat capacity of 20

J/K is heated irom 300 K to 400 K. calculate the change in cntropy of the gas.

3. (a)What is standerd enthalpy of formation? (b) Consider the formation of carbon dioxide from graphite

and oxygen: C(s) + O(g) → CO₂. Given that C(s) + O₂(g) → CO(g) (ΔH=-110.5 kJ/mol) and CO + O₂ → CO₂ (ΔH=-283.05 kJ/mol); Calculate the enthalpy of formation of carbon dioxide from graphics and oxygen.

4. (a)What is the relationship between equilibrium and entropy? Expla'a with matheinaticat equations).

(b)Calculate the standard reaction enthalpy for the combustion of methane, given thae the sta enthalpies of formation of methane, carbon (IV) oxide, and water vapour are -74.81, -393.5 sud -2 kJ/mol respectively.

(c) Three moles ef an ideal gas (Cy= 5 cal deg⁻¹mol⁻¹) et 10.0 atm and 0° are converted to 2.0 and 50°. Find

Ar ard A! for the change (R=2 cul mol 'deg⁻¹)

nsive and extensive propertics.

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5. (a) i. What do you unders'and by 'spon'ancous vicces.*? Give at feast three examples of spomtancaus

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ii, Explain why the change in entropy of a system is not always a suitable criterion for determining a spontaneous change?

(b) i. Give the mathematical expressions for the energy change, ΔH and the Gibbs free energy change ΔG .

energy change ΔG .

ii. Why is the Helmholtz free energy change, ΔA more significant in pressure-volume work systems than the Gibbs free energy change ΔG ?

(c) Calculate ΔG° at $5.00 \times 10^3 \text{ K}$ given that the entropy and enthalpy changes at this temperature are

$-1.25 \times 10^3 \text{ J mol}^{-1}$ and $-1.03 \times 10^4 \text{ J mol}^{-1}$ respectively. Is the reaction spontaneous?

» 20 (g); under what condition(s) would it be

6. (a). Consider the endothermic reaction: $\text{O}_2(\text{g})$

spontaneous? Explain the reason for your answer

(b). Comment on the spontaneity of a reaction if:

i. ΔH and ΔS are both positive iii. ΔH is negative and ΔS is positive

ii. ΔH and ΔS are both negative iv. ΔH is positive and ΔS is negative

(c). The decrease in Gibbs free energy and decrease in Helmholtz free energy may be expressed as

$-\Delta G = dW_{\text{ne}}$ and $-\Delta A = dW_{\text{re}}$ respectively, what is the physical significance of both?

(a) State the Maxwell's relations and their significance?

(b) The free energy change ΔG for a particular process is $-11.00 \text{ kJ mol}^{-1}$ at 293 K and $-102.00 \text{ J mol}^{-1}$ at 307 K . Using the Gibbs-Helmholtz equation, calculate the enthalpy change for the process at 300 K .

Paper 2 of 3 — CHM 223

Level 200 • Second Semester • PastQuestion

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DEPARTMENT OF CHEMISTRY
 COLLEGE OF SCIENCE
 FEDERAL UNIVERSITY OF PETROLEUM RESOURCES EFFURUN, NIGERIA
 SECOND SEMESTER EXAMINATIONS
 CHM 223: PHYSICAL CHEMISTRY II
 INSTRUCTIONS: ANSWER QUESTION ONE AND ANY OTHER TWO QUESTIONS
 UNIT: 2 CHEMISTS TIME: 2 HRS
 (1 atm = $1.0135 \times 10^5 \text{ Nm}^{-2}$)

- (a) Show that the decrease in Helmholtz free energy is equal to the maximum reversible work which a system can do isothermally.

(b) Calculate ΔA and ΔF when one mole of an ideal gas is allowed to expand isothermally at 300 K from a pressure of 100 atm to 1 atm.

(c) One mole of an ideal gas at 30°C is allowed to expand reversibly at constant temperature from volume 20 litres to 30 litres. Calculate the work done by gas in joules and calories?

(d) How many calories of heat are required to heat 16 g of Argon from 50°C to 110°C at (i) constant volume and (ii) constant pressure. Molar heat capacity of Argon at constant volume is 3 cal/degree and molar heat capacity of Argon at constant pressure is 5 cal/degree.
- (a) Show that the decrease in Gibbs free energy is equal to the reversible maximum work which a system can do isothermally and isobarically over and above the mechanical work.

(b) State the conditions for phase equilibrium.

(c) Draw a one-component phase diagram.
- (a) Explain the five types of thermodynamic processes.

(b) State four differences between Reversible and Irreversible processes.

(c) Calculate the energy in joules required to increase the volume of a substance by 1 cm³ against a pressure of 1 atm.
- (a) Thermodynamics is concerned only with equilibrium state, what are the criteria needed for a system to be in equilibrium.

(b) Find ΔE , q and w if 3 moles of hydrogen at 2 atm pressure expand isothermally at 60°C and reversibly to a pressure of 1 atm.

(c) Calculate the maximum work done when a pressure on 20 g of hydrogen is reduced from 30 atm to 1 atm at a constant temperature of 273 K. The gas behaves ideally. Will there be any change in internal energy? Also, calculate 'q'.
- (a) State the second and third law of thermodynamics.

(b) Explain the standard enthalpy of vaporization, standard enthalpy of fusion and standard enthalpy of sublimation.

(c) What is the molar enthalpy change for a temperature increase from 273 K to 355 K for Pb(s)? Given data:

Substance	a (J mol ⁻¹ K ⁻¹)	b (J mol ⁻¹ K ⁻²)	c (J mol ⁻¹ K ⁻¹)
C2g	16.86	8.77×10^{-5}	-8.54×10^4
C2l	44.22	8.79×10^{-5}	-8.54×10^4
H2O(l)	75.29	0	0
H2O(g)	28.58	8.79×10^{-5}	-3.76×10^4
Pb(s)	22.13	1.72×10^{-5}	5.69×10^4

H_2
 $1 \text{ mol} = 4.8145$
 $4.8145 = 1 \text{ kcal}$

Paper 3 of 3 — CHM 223 PHYSICAL CHEMISTRY

Level 200 • First Semester • Handout

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CHM 223 PHYSICAL CHEMISTRY II (2 CREDITS)

Chemical Thermodynamics

Introduction & Background. Le Chatelier's Principle. Systems and their surroundings. Work and heat as currencies of energy. **The First Law:** The conservation of energy: Internal energy. State functions and exact differentials. Expansion work. Reversible and irreversible changes. Heat capacity. Enthalpy. **Thermochemistry:** The standard state. Standard enthalpy of a phase transition. Standard reaction enthalpy. Standard enthalpy of formation. Temperature dependence of reaction enthalpy. **The Second Law:** The direction of spontaneous change: Entropy. The condition for equilibrium. Entropy of a phase transition. The variation of entropy with temperature and pressure. The third law. Statistical interpretation of entropy. **Free Energy:** The direction of spontaneous change: Helmholtz and Gibbs free energies. Available work. Fundamental equations and Maxwell relations. Variation of Gibbs free energy with temperature and pressure. **Phase Equilibria:** The condition for phase equilibrium. Clapeyron and Clausius-Clapeyron equations. One-component phase diagrams. **Chemical Equilibria:** Variation of Gibbs free energy with composition: The chemical potential of a species in a mixture. The extent of reaction. The condition for chemical equilibrium. Gibbs free energy of formation. Gibbs free energy of reaction. The reaction quotient. The equilibrium constant and its temperature dependence.

Le Châtelier's principle states that if a dynamic equilibrium is disturbed by changing the conditions, the position of equilibrium shifts to counteract the change to reestablish an equilibrium. If a chemical reaction is at equilibrium and experiences a change in pressure, temperature, or concentration of products or reactants, the equilibrium shifts in the opposite direction to offset the change.

Effect of temperature on Equilibrium

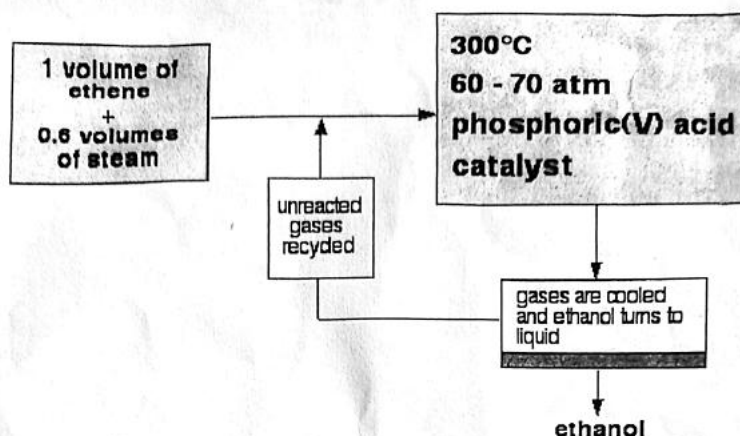
A temperature change occurs when temperature is increased or decreased by the flow of heat. This shifts chemical equilibria toward the products or reactants, which can be determined by studying the reaction and deciding whether it is endothermic or exothermic.

Manufacture of ethanol

Ethanol is manufactured by reacting ethene with steam. The reaction is reversible, and the formation of the ethanol is exothermic.



Only 5% of the ethene is converted into ethanol at each pass through the reactor. By removing the ethanol from the equilibrium mixture and recycling the ethene, it is possible to achieve an overall 95% conversion. A flow scheme for the reaction looks like this:



The proportions of ethene and steam

The equation shows that the ethene and steam react 1 : 1. In order to get this ratio, you would have to use equal volumes of the two gases.



Because water is cheap, it would seem sensible to use an excess of steam in order to move the position of equilibrium to the right according to Le Chatelier's Principle. In practice, an excess of ethene is used. This is very surprising at first sight. Even if the reaction was one-way, you couldn't possibly convert all the ethene into ethanol. There isn't enough steam to react with it.

The reason for this oddity lies with the nature of the catalyst. The catalyst is phosphoric(V) acid coated onto a solid silicon dioxide support. If you use too much steam, it dilutes the catalyst and can even wash it off the support, making it useless.

The temperature

You need to shift the position of the equilibrium as far as possible to the right in order to produce the maximum possible amount of ethanol in the equilibrium mixture. The forward reaction (the production of ethanol) is exothermic.

According to Le Chatelier's Principle, this will be favoured if you lower the temperature. The system will respond by moving the position of equilibrium to counteract this - in other words by producing more heat. In order to get as much ethanol as possible in the equilibrium mixture, you need as low a temperature as possible. However, 300°C isn't particularly low.

Rate considerations

The lower the temperature you use, the slower the reaction becomes. A manufacturer is trying to produce as much ethanol as possible per day. It makes no sense to try to achieve an equilibrium mixture which contains a very high proportion of ethanol if it takes several years for the reaction to reach that equilibrium. You need the gases to reach equilibrium within the very short time that they will be in contact with the catalyst in the reactor.

The compromise

300°C is a compromise temperature producing an acceptable proportion of ethanol in the equilibrium mixture, but in a very short time. Under these conditions, about 5% of the ethene reacts to give ethanol at each pass over the catalyst.

The pressure

Equilibrium considerations /

Notice that there are 2 molecules on the left-hand side of the equation, but only 1 on the right. According to Le Chatelier's Principle, if you increase the pressure the system will respond by favoring the reaction which produces fewer molecules. That will cause the pressure to fall again. In order to get as much ethanol as possible in the equilibrium mixture, you need as high a pressure as possible. High pressures also increase the rate of the reaction. However, the pressure used isn't all that high.

Problems with high pressures

There are two quite separate problems in this case:

- High pressures are expensive. It costs more to build the original plant because you need extremely strong pipes and containment vessels. It also needs a lot of energy to produce the high pressures. That can make the ethanol uneconomic to produce.
- At high pressures, the ethene polymerizes to make poly(ethene). Apart from wasting ethene, this could also clog up the plant.

The catalyst

The catalyst has no effect whatsoever on the position of the equilibrium. Adding a catalyst doesn't produce any greater percentage of ethanol in the equilibrium mixture. Its only function is to speed up the reaction.

Rate considerations

In the absence of a catalyst the reaction is so slow that virtually no reaction happens in any sensible time. The catalyst ensures that the reaction is fast enough for a dynamic equilibrium to be set up within the very short time that the gases are actually in the reactor.